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Printing apparatus receives print job from external apparatus without holding the print job in accordance with job attributes, print control method, and non-transitory computer-readable storage medium

Abstract

A cloud print service specifies a job release method as a job attribute. In accordance with the job attribute, a printing apparatus controls whether a secure printing is applied or printing is to be performed without holding the print job.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a printing apparatus in a cloud print service, a control method, and a storage medium therefor.

Description of the Related Art

(2) In conventional print services, a user directly transmits a print job from a client terminal, such as a PC, to a printing apparatus to print the print job. On the other hand, in recent years, cloud print services using a cloud service on the Internet have been proposed.

(3) A conventionally known processing sequence of cloud print will be described below with reference to FIG. 1. A printing apparatus **101** supports a cloud print function and a Web user interface (UI) function for an operation of the printing apparatus **101**. A client terminal **100** supports a cloud print client function and a Web UI client for an operation of the Web UI. A cloud print service **102** represents a cloud print service on the Internet.

(4) The user transmits a cloud print service registration request **111** to the printing apparatus **101** via the Web UI using the Web UI client of the client terminal **100**. Upon receiving the cloud print

service registration request **111**, the printing apparatus **101** transmits a cloud print service registration request **112** to the cloud print service **102**. Upon receiving the cloud print service registration request **112**, the cloud print service **102** transmits a cloud print service registration request response **113** containing a registration URL **130** for cloud print registration to the printing apparatus **101**. Upon receiving the cloud print service registration request response **113**, the printing apparatus **101** transmits cloud print service registration URL display screen information **114** containing the registration URL **130** to the client terminal **100** via the Web UI.

(5) Upon receiving the cloud print service registration URL display screen information **114**, the client terminal **100** displays the registration URL **130** on the Web UI client. The user operates the Web UI client of the client terminal **100** to access the registration URL **130** displayed on the Web UI client. In response to the user's operation, the client terminal **100** transmits a cloud print service registration approval request **115** containing a cloud account **131** to the cloud print service **102**. Upon receiving the cloud print service registration approval request **115**, the cloud print service **102** transmits a cloud print service registration approval response **116** to the client terminal **100**. The cloud account **131** indicates that the user has permission to use the cloud print service **102**, and the user has registered a user account in the cloud print service **102** in advance.

(6) The printing apparatus **101** transmits a cloud print service registration verification request **117** to the cloud print service **102**. Upon receiving the cloud print service registration verification request **117**, the cloud print service **102** transmits a cloud print service registration verification response **118** containing a cloud printer ID **132** to the printing apparatus **101**. In this processing, registration of the printing apparatus **101** in the cloud print service **102** is completed, whereby the printing apparatus **101** is able to use the cloud print service **102**.

(7) After completion of the registration by the above described procedure, the printing apparatus **101** transmits a cloud print service printer information update request **140** to the cloud print service **102**. The cloud print service printer information update request **140** contains a printer attribute **135** of the printing apparatus **101**. Upon receiving the cloud print service printer information update request **140**, the cloud print service **102** transmits a cloud print service printer information update response **141** to the printing apparatus **101**.

(8) Then, the printing apparatus **101** transmits an event request **123** to the cloud print service **102**. In the event request **123**, a job reception event is specified as a type of requesting event.

(9) The client terminal **100** transmits a cloud printer information request **142** to the cloud print service **102**. The cloud printer information request **142** contains the cloud account **131**. Upon receiving the cloud printer information request **142**, the cloud print service **102** acquires printer information about a printer available to the cloud account **131** from among printers contained in the cloud printer information request **142**. Then, a cloud printer information response **143** containing the cloud printer ID **132** and the printer attribute **135** serving as the printer information is transmitted to the client terminal **100**. The printer attribute **135** has been transmitted from the printing apparatus **101** to the cloud print service **102** in response to the cloud print service printer information update request **140**.

(10) The client terminal **100** transmits a print request **119** containing the cloud printer ID **132**, print data **134**, and the cloud account **131** to the cloud print service **102**.

(11) Upon receiving the print request **119**, the cloud print service **102** transmits an event response **124** to the printing apparatus **101**. The event response **124** is transmitted as a response to the event request **123**, and a job reception event is specified as a type of the event.

(12) Upon receiving the event response **124**, the printing apparatus **101** transmits a print job request **120** to the cloud print service **102**. The cloud print service **102** transmits a print job response **121** containing the print data **134** and the cloud account **131** to the printing apparatus **101**. The printing apparatus **101** having received the print data **134** prints the print data **134**. The above described operation is also discussed in Japanese Patent Application Laid-Open No. 2021-170258.

(13) Japanese Patent Application Laid-Open No. 2020-009062 discusses hold printing in which a

print job transmitted from a user without designating a printing apparatus is temporarily held in a print server, and printing is executed in response to a new print instruction from the user in a state in which a print target printing apparatus is designated.

SUMMARY OF THE INVENTION

(14) According to an aspect of the present invention, a printing apparatus configured to receive a first print job from an external apparatus and print the first print job, the printing apparatus includes a setting unit configured to enable a secure print setting with which the first print job received from the external apparatus is to be held without being printed, and the held print job is printed in response to an instruction from a user to print the print job, and a print control unit configured to receive a second print job from a print service via a network and printing the second print job, wherein even in a case where the secure print setting is enabled by the setting unit, the print control unit prints the second print job received from the print service without holding the print job in accordance with a job attribute of the received print job.

(15) Further features of the present invention will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a sequence diagram illustrating cloud print processing.

(2) FIG. 2 is a diagram illustrating a network configuration of pull printing in cloud print.

(3) FIG. 3 is a diagram illustrating a network configuration of secure print.

(4) FIG. 4 is a diagram illustrating a hardware configuration of a printing apparatus according to present embodiments.

(5) FIG. 5 is a diagram illustrating a software configuration of the printing apparatus according to the present embodiments.

(6) FIG. 6 is a diagram illustrating an operation panel of the printing apparatus according to the present embodiments.

(7) FIG. 7 is a diagram illustrating a hardware configuration of a cloud print server according to the present embodiments.

(8) FIG. 8 is a diagram illustrating a software configuration diagram of the cloud print server according to the present embodiments.

(9) FIG. 9 is a sequence diagram illustrating processing that is executed by the printing apparatus according to a first embodiment when pull printing in cloud print is performed.

(10) FIG. 10 is a diagram illustrating attribute information specifying a job release method that is supported by the printing apparatus in the first embodiment.

(11) FIG. 11 is a diagram illustrating a screen on which a secure print setting is set in the printing apparatus according to the first embodiment.

(12) FIG. 12 is a flowchart illustrating processing that is performed by the printing apparatus according to the first embodiment in response to receipt of a cloud print job reception event.

(13) FIG. 13 is a flowchart illustrating processing that is performed by the cloud print server according to the first embodiment in response to receipt of a print request of a job.

(14) FIG. 14 is a diagram illustrating a network configuration of pull printing in cloud print according to a second embodiment.

(15) FIG. 15 is a flowchart illustrating processing that is performed by the printing apparatus according to the second embodiment in response to receipt of a cloud print job reception event.

(16) FIG. 16 is a diagram illustrating attribute information specifying a job release method that is supported by the printing apparatus according to the second embodiment.

(17) FIG. 17 is a diagram illustrating a printing apparatus management screen of the cloud print

service according to the second embodiment.

(18) FIG. 18 is a flowchart illustrating processing that is performed by the cloud print server according to the second embodiment in response to receipt of a print request of a job.

DESCRIPTION OF THE EMBODIMENTS

(19) In addition to normal hold printing, in a case where cloud print discussed in Japanese Patent Application Laid-Open No. 2021-170258 supports hold printing, two types of hold printing are performed by a printing apparatus. This may lead to increase in the time and effort of an operation of a user who operates the printing apparatus. Therefore, the present invention is directed to providing a method for reducing the time and effort of a user's operation in a case where cloud print supports hold printing. Each of the embodiments of the present invention described below can be implemented solely or as a combination of a plurality of the embodiments or features thereof where necessary or where the combination of elements or features from individual embodiments in a single embodiment is beneficial.

(20) First, two types of hold printing in the present embodiments will be described.

(21) FIG. 2 is a diagram illustrating a network configuration of pull printing in cloud print. In the cloud print service 102, there are registered printers 210 in one to-one correspondence with printing apparatuses 101. Each of the registered printers 210 reflects a status of the corresponding one of the printing apparatuses 101 and a status of a job in progress in the corresponding printing apparatus. In the present embodiment, a printing apparatus 201 corresponds to a registered printer 211, a printing apparatus 202 corresponds to a registered printer 212, and a printing apparatus 203 corresponds to a registered printer 213. A mobile terminal 204 is able to display a list of print jobs of a logged-in user. The displaying is realized by step 251 in which the mobile terminal 204 acquires a print job of a user who has logged in from the cloud print service 102.

(22) On the mobile terminal 204, the user issues a print instruction to the cloud print service 102 by designating the printing apparatus 101 and a print job. This processing is realized by step 252 in which the mobile terminal 204 transmits the print job and information on the printing apparatus 101 to the cloud print service 102.

(23) The cloud print service 102 has a virtual printer 220. A plurality of registered printers 210 can be associated with the virtual printer 220. In the present embodiment, the registered printers 211, 212, and 213 are associated with the virtual printer 220.

(24) A processing procedure of pull printing using the virtual printer 220 will be described below. In step 231, the user registers a print job in the virtual printer 220 from the client terminal 100. In step 232, the user logs in to the mobile terminal 204 and displays a print job list of the user. In step 233, the user designates a desired print job and issues a print instruction. In step 234, the printing apparatus 201 requests the corresponding printer, which is registered printer 211, to transmit the print job designated in step 233.

(25) In step 235, the registered printer 211 requests the print job from the virtual printer 220. In step 236, the virtual printer 220 transfers a print job 240 to the registered printer 211. In step 237, the registered printer 211 transmits the print job 240 to the printing apparatus 201 as a response to the print job request in step 234. The printing apparatus 201 having received the print job 240 prints the print job 240.

(26) With the above-described procedure, the user is able to print a print job input to the virtual printer 220 by using any printing apparatus 101 associated with the virtual printer 220.

(27) Secure print, which is another type of hold printing, will be described below. FIG. 3 is a diagram illustrating a network configuration of secure print in the printing apparatus 101. In secure print, upon receiving a print job, the printing apparatus 101 stores the print job in a storage 310, for example, an HDD, without executing printing. In step 351, when the user performs printing, the user logs in to the printing apparatus 101 by using an operation panel. Then, in step 352, the user refers to a print job list of the user, designates a desired print job, and issues a print instruction to perform printing.

(28) A method to be used by the printing apparatus **101** in the above described processing to receive the print job can be any method. FIG. 3 illustrates cases in which a client terminal **320** transmits a print job to the printing apparatus **101** by using the Line Printer Remote (LPR) protocol in step **301**, a client terminal **321** transmits a print job to the printing apparatus **101** by using the RAW protocol in step **302**, a client terminal **322** transmits a print job to the printing apparatus **101** by using the Internet Printing Protocol (IPP) in step **303**, and the client terminal **100** transmits a print job to the printing apparatus **101** via cloud print in steps **304** and **305**. A print instruction for the secure print is issued twice, that is, at a timing of step **304** when the user instructs to print data to be printed and at a timing of step **352** when the user instructs to print the held print job.

(29) In a case where the pull printing in the cloud print described with reference to FIG. 2 and the secure print described with reference to FIG. 3 are both enabled, the user performs a login operation twice to perform printing, which deteriorates operability. Specifically, in order to perform the pull printing in the cloud print, the user logs in to the mobile terminal **204** and displays the print job list of the user in step **232**. Then, in step **233**, the user designates a desired job and issues a print instruction. In response to the print instruction, the print job is stored in the storage **310** of the printing apparatus **101**. The user logs in to the printing apparatus **101** by using the operation panel, refers to the print job list of the user, designates a desired print job, and performs printing.

(30) Hereinafter, in the present invention, the mode for addressing the above described issue will be described with reference to the drawings. The following embodiments are not intended to limit the present invention according to the appended claims, and not all combinations of features described in the embodiments are necessarily essential to the solving means of the present invention.

(31) First, FIG. 4 is a block diagram illustrating a hardware configuration of the printing apparatus **101** according to the present invention. A control unit **401** including a central processing unit (CPU) **402** controls entire operation of the printing apparatus **101**. The CPU **402** reads a control program stored in a read-only memory (ROM) **404** and performs various kinds of control, such as communication control. A random access memory (RAM) **403** is used as, for example, a main memory and a temporary storage area of a work area of the CPU **402**. A hard disc drive (HDD) **414** stores data, various programs, and various information tables, for example.

(32) The printer OF **406** serves as an interface to output an image signal to a printer **407** (printer engine). A scanner OF **408** serves as an interface to input a read image signal from a scanner **409** (scanner engine). The CPU **402** processes an image signal input from the scanner OF **408**, and outputs the processed image signal to the printer OF **406** as a recording image signal.

(33) The operation panel I/F **410** connects an operation panel **411** with the control unit **401**. The operation panel **411** includes, for example, a liquid crystal display (LCD) unit having a touch panel function and a keyboard.

(34) The network OF **412** transmits information to an external terminal, such as the client terminal **100** and the cloud print service **102**, and receives various types of information from the external terminal.

(35) The blocks in the control unit **401** are connected with each other via a system bus **405**.

(36) FIG. 5 is a diagram illustrating a software configuration of the printing apparatus **101**. Each functional unit in FIG. 5 is realized by the CPU **402**, which is included in the printing apparatus **101**, executing a control program.

(37) An operation control unit **500** controls the operation panel **411**. The operation control unit **500** displays an operation menu on the operation panel **411**, waits for an instruction input from the user, notifies other functional units of a received instruction content, and displays a result of the instruction on the operation panel **411**.

(38) The cloud print service registration control unit **501** analyzes the cloud print service registration request **111** received from the client terminal **100**, and transmits the cloud print service registration request **112** to the cloud print service **102**. This processing controls a cloud print service registration process.

- (39) The cloud print printing control unit **502** transfers the print data **134** received from the cloud print service **102** to an image processing unit **507**. The image processing unit **507** performs processing for rendering a print job into image data for printing.
- (40) A print processing unit **508** performs processing for printing the image data obtained by rendering performed by the image processing unit **507**.
- (41) A cloud print job information management unit **503** manages job information on the printing apparatus **101**, and notifies the cloud print service **102** of the job information.
- (42) A cloud print event control unit **504** controls event transmission and event reception processing between the printing apparatus **101** and the cloud print service **102**.
- (43) A cloud print apparatus information management unit **505** manages apparatus information on the printing apparatus **101** and notifies the cloud print service **102** of the apparatus information.
- (44) A storage unit **506** stores, in response to an instruction from other functional units, designated data in the ROM **404** or the HDD **414**, or reads out stored data. An example of data managed by the storage unit **506** is print job information.
- (45) A secure print job control unit **509** controls a secure print job.
- (46) FIG. **6** is a diagram illustrating the operation panel **411** of the printing apparatus **101**. A display panel **605** is an LCD, for example. A button **601** is pressed when copying using the printing apparatus **101** is performed, and pressing the button **601** displays a copy operation screen on the display panel **605**. A button **602** is pressed when faxing using the printing apparatus **101** is performed, and pressing the button **602** displays a fax operation screen on the display panel **605**. A button **603** is pressed when scanning using the printing apparatus **101** is performed, and pressing the button **603** displays a scan operation screen on the display panel **605**. The display panel **605** also displays a dialog to communicate with the user. In the present embodiment, the display panel **605** further displays a user login screen and a print job list display screen.
- (47) A numeric keypad **608** is used to input, for example, numerals. An OK key **607** is used when a display content on the display panel **605** is determined. A direction key **606** is used to select a menu displayed on the display panel **605**. A button **609** is used to perform monochrome copy, and a button **610** is used to perform color copy. A button **611** is used to stop processing. A button **612** is used to display a menu screen on which a setting of the printing apparatus **101** is set. A button **613** is used to check the status of the printing apparatus **101**, for example, a list of print jobs received by the printing apparatus **101**.
- (48) FIG. **7** is a block diagram illustrating a hardware configuration of a cloud server on which the cloud print service **102** according to the present invention is executed. A control unit **701** including a CPU **702** controls entire operation of the cloud print service **102**. The CPU **702** reads out a control program stored in a ROM **704** and performs various kinds of control, such as communication control. A RAM **703** is a temporary storage area serving as, for example, a main memory or a work area of the CPU **702**. An HDD **705** stores data, various programs, and various information tables, for example. The network OF **710** transmits information to an external terminal, such as the client terminal **100** and the printing apparatus **101**, and receives various kinds of information from the external terminal.
- (49) The blocks in the control unit **701** are connected with each other via a system bus **706**.
- (50) FIG. **8** is a diagram illustrating a software configuration of a cloud server on which the cloud print service **102** is executed. Each functional unit illustrated in FIG. **8** is implemented by the CPU **702** included in the cloud server executing a control program.
- (51) A cloud print service registration control unit **801** controls processing for receiving the cloud print service registration request **112** from the printing apparatus **101** and registering the printing apparatus **101** in the cloud print service **102**. A cloud print printing control unit **802** receives the print request **119** from the client terminal **100**, and controls processing for transmitting the print data **134** contained in the print request **119** to the printing apparatus **101**. A cloud print event control unit **803** receives the event request **123** from the printing apparatus **101**, and controls

processing for transmitting the event response **124** in accordance with whether an event has occurred.

(52) A cloud print job information management unit **806** manages information on a print job and a status of the print job.

(53) A cloud printer management unit **804** manages a plurality of printing apparatuses **101** registered in the cloud print service **102**. More specifically, the cloud printer management unit **804** stores and manages, for each printing apparatus **101**, printing apparatus capability information and information on a print queue for which a print job is spooled.

(54) In response to an instruction from other functional units, a storage unit **805** stores designated data in the ROM **404** or the HDD **414**, or reads out the stored data. Examples of data managed by the storage unit **805** include the print data **134**, event information, and capability information on the printing apparatus **101** registered in the cloud print service **102**.

(55) A virtual printer control unit **807** manages information on the virtual printer **220** registered in the cloud print service **102**. The virtual printer **220** is a conceptual printer object existing on the cloud print service **102**. The virtual printer **220** does not directly correspond to the printing apparatus **101** which is a real printer. The information on the virtual printer **220** includes, for example, capability information on the virtual printer **220**, a print queue, and information on the registered printer **210** associated with the virtual printer **220**.

(56) A registered printer control unit **808** manages information on the registered printer **210** registered in the cloud print service **102**. The registered printer **210** is a conceptual printer object existing on the cloud print service **102**. The registered printer **210** is generated when the printing apparatus **101** is registered in the cloud print service **102**, and has a one-to-one correspondence with the printing apparatus **101** registered in the cloud print service **102**. The information on the registered printer **210** includes, for example, capability information on the registered printer **210** and information on a print queue.

(57) In the following description, the printing apparatus **101** and the cloud print service **102** communicate with each other using IPP.

(58) Hereinafter, a first embodiment and a second embodiment will be described based on the above-described configuration.

(59) FIG. **9** is a sequence diagram illustrating processing that is executed by the printing apparatus **101** according to the first embodiment when pull printing in the cloud print is performed.

(60) In step **901**, the user logs in to the mobile terminal **204**. In step **902** the user designates a desired printing apparatus **101** and performs an operation for displaying a print job list of the user.

(61) In step **903**, the mobile terminal **204** requests the registered printer **210** on the cloud print service **102** to send the print job list of the user who has logged in in step **901**.

(62) In step **904**, the registered printer **210** on the cloud print service **102** requests the print job list of the user designated in step **903** from the virtual printer **220**.

(63) In step **905**, the virtual printer **220** on the cloud print service **102** responds to the registered printer **210** with the print job list of the user designated in step **904**.

(64) In step **906**, the registered printer **210** on the cloud print service **102** responds to the mobile terminal **204** with the print job list of the user designated in step **903**.

(65) In step **907**, the mobile terminal **204** displays the received print job list.

(66) In step **908**, the user designates a desired job in the displayed print job list and performs an operation for performing printing.

(67) In step **909**, the mobile terminal **204** transmits a print request for the designated print job to the registered printer **210** on the cloud print service **102**.

(68) In step **910**, the registered printer **210** on the cloud print service **102** requests the virtual printer **220** to transmit information about the print job designated in step **909**.

(69) In step **911**, the virtual printer **220** on the cloud print service **102** responds to the registered printer **210** with print job information designated in step **910**.

(70) In step **912**, the registered printer **210** on the cloud print service **102** responds to the printing apparatus **101** with a job reception event. Specifically, the registered printer **210** transmits a response for “Get-Notifications”, which is an IPP request of event acquisition. In the response, “job-fetchable” indicating the presence of a received job is specified as the event type.

(71) In step **913**, the printing apparatus **101** transmits a print job list request to the registered printer **210** on the cloud print service **102**. Specifically, the printing apparatus **101** transmits a request of “Get-Jobs”, which is an IPP request of print job list acquisition.

(72) In step **914**, the registered printer **210** on the cloud print service **102** transmits a job list response to the printing apparatus **101**. Specifically, the cloud print service **102** transmits a response for “Get-Jobs”, which is an IPP request of event acquisition.

(73) In step **915**, the printing apparatus **101** transmits a job attribute request to the registered printer **210** on the cloud print service **102**. Specifically, the printing apparatus **101** transmits a request of “Fetch-Job”, which is an IPP request of job attribute acquisition. In this processing, “job-id” acquired in step **914** is used to specify “job-id”.

(74) In step **916**, the registered printer **210** on the cloud print service **102** transmits a job attribute response to the printing apparatus **101**. Specifically, the registered printer **210** transmits a response for “Fetch-Job”, which is an IPP request of job attribute acquisition. In this processing, “job-release-method” specifying a job release method is set to the job attribute.

(75) In step **917**, the printing apparatus **101** transmits a job-data request to the registered printer **210** on the cloud print service **102**. Specifically, the printing apparatus **101** transmits a request of “Fetch-Document”, which is an IPP request of job data acquisition. In this processing, “job-id” acquired in step **914** is used to specify “job-id”.

(76) In step **918**, the registered printer **210** on the cloud print service **102** transmits a job data response to the printing apparatus **101**. Specifically, the registered printer **210** transmits a response for “Fetch-Document”, which is an IPP request of job data acquisition.

(77) In step **919**, the printing apparatus **101** outputs the received job in accordance with the received job attribute.

(78) FIG. **10** is a diagram illustrating attribute information with which a job release method that is supported by the printing apparatus according to the first embodiment is specified.

(79) The attribute information is notified as a print attribute from the cloud print service **102** to the printing apparatus **101** in step **916** of FIG. **9**. Thus, in a case of a print job transmitted from an external apparatus different from the cloud print service **102**, the print job does not contain the job attribute.

(80) The job attribute is able to be specified by the cloud print service **102**.

(81) As shown in an item **1001**, “job-release-method” is defined as the job attribute.

(82) The job attribute “job-release-method” supports a value “system-specified” and a value “forced-print” as shown in items **1002** and **1003**.

(83) In a case where “system-specified” in the item **1002** is specified, the printing apparatus **101** performs processing in accordance with a release method set in the printing apparatus **101**.

(84) In a case where “forced-print” in the item **1003** is specified, the printing apparatus **101** does not hold the print job after receiving the print job, regardless of the release method set in the printing apparatus, and executes printing without a print instruction from the user.

(85) FIG. **11** is a diagram illustrating a screen on which a secure print setting is performed in the printing apparatus **101** according to the first embodiment.

(86) The screen is displayed on the display panel **605** of the printing apparatus **101**. A print setting screen **1101** is a top screen of a print setting of the printing apparatus **101**. A field **1102** displays a setting of the secure print. A field **1103** displays a retention time of a job stored in the printing apparatus **101**.

(87) In response to the field **1102** being touched, the screen transitions to a screen **1111** on which the secure print setting is changed. On the screen **1111**, the secure print setting is able to be

switched between ON and OFF.

(88) In a case where the secure print setting is set to ON, the printing apparatus **101** stores all received print jobs in the storage **310**. Printing is not performed at the timing of print job reception, and printing is performed when a print instruction for the print job is newly received from the user. Such a printing method is referred to as hold print.

(89) FIG. **12** is a flowchart illustrating processing that is performed by the printing apparatus **101** according to the first embodiment receiving a cloud print job reception event.

(90) In step **S1201**, the cloud print printing control unit **502** of the printing apparatus **101** acquires a print job list by sending an inquiry to the cloud print service **102**. After acquisition of the print job list, processing from step **S1202** to step **S1211** is repeated for the number of jobs.

(91) In step **S1203**, the cloud print printing control unit **502** of the printing apparatus **101** acquires a job attribute of a designated job by sending an inquiry to the cloud print service **102**.

(92) In step **S1204**, the cloud print printing control unit **502** of the printing apparatus **101** acquires job data of the designated job by sending an inquiry to the cloud print service **102**.

(93) In step **S1205**, the cloud print job information management unit **503** of the printing apparatus **101** acquires a value of “job-release-method” from the print attribute acquired in step **S1203**.

(94) In step **S1206**, the cloud print printing control unit **502** of the printing apparatus **101** changes the subsequent processing according to the value of “job-release-method”. In a case where the value of “job-release-method” is “forced-print” (forced-print in step **1206**), the processing proceeds to step **S1209**. In step **S1209**, the cloud print printing control unit **502** of the printing apparatus **101** prints the acquired job data in accordance with the acquired job attribute. In a case where the value of “job-release-method” is “system-specified” or in a case where the job attribute does not exist (system-specified or NO ATTRIBUTE in step **1206**), the processing proceeds to step **S1207**. In **S1207**, the storage unit **506** of the printing apparatus **101** reads out the secure print setting stored in HDD **414**.

(95) In step **S1208**, the cloud print printing control unit **502** of the printing apparatus **101** changes the subsequent processing according to a value of the secure print setting. In a case where the value of the secure print setting is OFF (YES in step **S1207**), the processing proceeds to step **S1209**. In step **S1209**, the cloud print printing control unit **502** of the printing apparatus **101** prints the acquired job data in accordance with the acquired job attribute. In a case where the value of the secure print setting is ON (NO in step **S1207**), the processing is proceeds to step **S1210**. In step **S1210**, the storage unit **506** of the printing apparatus **101** stores the acquired job data in the HDD **414**.

(96) FIG. **13** is a flowchart illustrating processing that is performed by the cloud print server according to the first embodiment in response to receipt of a print request of a job.

(97) In step **S1301**, the cloud print printing control unit **802** of the cloud print server acquires information about a printing apparatus and information about a print job (job-id) which are contained in the print request.

(98) In step **S1302**, the cloud print printing control unit **802** of the cloud print server transfers the designated print job from the virtual printer **220** to the registered printer **210**.

(99) In step **S1303**, the cloud print printing control unit **802** of the cloud print server acquires transmission source information on the print request.

(100) In step **S1304**, the cloud print printing control unit **802** of the cloud print server changes the subsequent processing according to the transmission source information on the print request.

(101) In a case where the transmission source is a mobile terminal (YES in step **S1304**), the processing proceeds to step **S1306**. In step **S1306**, the cloud print job information management unit **806** of the cloud print server adds “job-release-method” to the job attribute of the target print job, and sets “forced-print” as the value.

(102) In a case where the transmission source is not a mobile device (NO in step **S1304**), the processing proceeds to step **S1305**. In step **S1305**, the cloud print job information management unit

806 of the cloud print server adds “job-release-method” to the job attribute of the target print job and sets “system-specified” as the value.

(103) In step **S1307**, the cloud print job information management unit **806** of the cloud print server sets “job-fetchable” as the value of the job attribute “job-state-reasons” of the target print job, whereby the print job is able to be acquired by the printing apparatus **101**.

(104) Next, a second embodiment will be described. In the first embodiment, the user performs a login operation once in the pull printing processing performed from the mobile terminal **204**.

(105) In the second embodiment, hold print is forcibly performed in a case where a print job is directly input to a registered printer, in a configuration realizing pull printing in cloud print. The redundant description of those similar to the first embodiment is omitted.

(106) FIG. **14** is a network diagram illustrating a configuration of pull printing in cloud print according to the second embodiment. Since the configuration illustrated in FIG. **14** is substantially the same as the configuration of the pull printing illustrated in FIG. **2**, the difference will be described, and the redundant description will be omitted.

(107) In FIG. **2**, the processing procedure of a case in which a print job is input from the client terminal **100** to the virtual printer **220** is illustrated. In FIG. **14**, a processing procedure of a case in which a print job is input from the client terminal **100** to the registered printer **211** is illustrated, which is the difference from the processing procedure illustrated FIG. **2**.

(108) A processing procedure for directly registering a print job in a registered printer in a pull printing configuration using the virtual printer **220** will be described below.

(109) In step **1401**, the user registers a print job in the registered printer **211** from the client terminal **100**. In step **1403**, the printing apparatus **201** requests a print job **1402** registered in step **1401** from the registered printer **211** corresponding to the printing apparatus **201**. In step **1404**, the registered printer **211** transmits the print job **1402** to the printing apparatus **201** as a response to the request for the print job in step **1403**. The printing apparatus **201** having received the print job **1402** prints the print job **1402**.

(110) With the above-described configuration, the user is able to perform printing using the printing apparatus **101** by directly designating the printing apparatus **101** to perform printing even in the case of the configuration of pull printing.

(111) FIG. **15** is a flowchart illustrating processing that is performed by the printing apparatus according to the second embodiment in response to receipt of a cloud print job reception event.

(112) In step **S1501**, the cloud print printing control unit **502** of the printing apparatus **101** acquires a print job list by sending an inquiry to the cloud print service **102**. After acquisition of the print job list, the processing from step **S1502** to step **S1511** is repeated for the number of jobs.

(113) In step **S1503**, the cloud print printing control unit **502** of the printing apparatus **101** acquires a job attribute of a job designated by sending an inquiry to the cloud print service **102**.

(114) In step **S1504**, the cloud print printing control unit **502** of the printing apparatus **101** acquires job data of the job designated by sending an inquiry to the cloud print service **102**.

(115) In step **S1505**, the cloud print job information management unit **503** of the printing apparatus **101** acquires a value of “job-release-method” from the job attribute acquired in step **S1503**.

(116) In step **S1506**, the cloud print printing control unit **502** of the printing apparatus **101** changes the subsequent processing according to the value of “job-release-method”. In a case where “forced-print” is set as the value of “job-release-method” (forced-print in step **S1506**), the processing proceeds to step **S1509**. In step **S1509**, the cloud print printing control unit **502** of the printing apparatus **101** prints the acquired job data in accordance with the acquired job attribute. In a case where “system-specified” is set as the value of “job-release-method” or in a case where the job attribute does not exist (system-specified or NO ATTRIBUTE in step **S1506**), the processing proceeds to step **S1507**. In step **S1507**, the storage unit **506** of the printing apparatus **101** reads out the secure print setting stored in the HDD **414**. In a case where “forced-store” is set as the value of “job-release-method” (forced-store in step **S1506**), the processing proceeds to step **S1510**. In step

S1510, the storage unit 506 of the printing apparatus 101 stores the acquired job data in HDD 414. (117) In step S1508, the cloud print printing control unit 502 of the printing apparatus 101 changes the subsequent processing according to a value of the secure print setting. In a case where the value of the secure print setting is OFF (YES in step S1508), the processing proceeds to step S1509. In step S1509, the cloud print printing control unit 502 of the printing apparatus 101 prints the acquired job data in accordance with the acquired job attribute. In a case where the value of the secure print setting is ON (NO in step S1508), the processing proceeds to step S1510. In step S1510, the storage unit 506 of the printing apparatus 101 stores the acquired job data in the HDD 414.

(118) FIG. 16 is a diagram illustrating attribute information specifying a job release method that is supported by the printing apparatus 101 according to the second embodiment. The method illustrated in FIG. 16 is substantially the same as the job release method, which is supported by the printing apparatus 101, described with reference to FIG. 10. The difference will be described, and the redundant description will be omitted.

(119) As the value of the job attribute “job-release-method” in the item 1001, “forced-store” in an item 1601 is able to be specified. In a case where “forced-store” in the item 1601 is specified, the printing apparatus 101 does not print the acquired job data regardless of the release method set in the printing apparatus 101, and stores the job data in the storage area.

(120) FIG. 17 is a diagram illustrating a printing apparatus management screen in the cloud print service 102 according to the second embodiment. In the cloud print service 102, registered cloud printers can be managed in an integrated manner, and a setting for controlling a cloud print operation can be set for each printing apparatus.

(121) A screen 1701 is a top screen for cloud printer management. A list of registered printers is displayed in a field 1704. In the present embodiment, a printer A in a field 1702 and a printer B in a field 1703 are registered. A selection of a displayed printer causes a transition to a screen 1711 on which a setting for the selected printer is performed.

(122) In the present embodiment, a description is given of a case in which the screen 1711 is displayed in response to the printer A being selected. On the screen 1711 for a printer setting, an installation location of the printer A is able to be set in a field 1712. In the present embodiment, the printer A is installed in a shared office A. On the screen 1711 for the printer setting, a print job output method is able to be set in a field 1713. As the output method of the printer A, the following three settings can be set.

(123) The first among the three settings is a setting “with printer setting”. With this setting, the output method is controlled in accordance with the printer setting. No special control is performed in the cloud print service 102.

(124) The second among the three settings is a setting “forced print”. With this setting, printing is forcibly performed in the printing apparatus 101 regardless of the printer setting. In the cloud print service 102, this control is realized by setting “forced-print” as the value of “job-release-method” which is a job attribute of a print job.

(125) The third among the three settings is a setting “forced store”. With this setting, storing of job data is forcibly performed in the printing apparatus 101 regardless of the printer setting. In the cloud print service 102, this control is realized by setting “forced-store” as the value of “job-release-method” which is a job attribute of a print job.

(126) In the present embodiment, since the setting “forced store” is selected, all jobs are stored in the printing apparatus 101. In the present embodiment, the printing apparatus 101 is installed in the shared office A, and the printed product may be seen by an unspecified number of people. In such a case, the setting “forced store” is useful from the viewpoint of security.

(127) FIG. 18 is a flowchart illustrating processing that is performed by the cloud print server according to the second embodiment in response to receipt of a print request of a print job.

(128) In step S1801, the cloud print printing control unit 802 of the cloud print server acquires

information on a printing apparatus and information on a print job (job IDs) which are contained in the print request.

(129) In step **S1802**, the cloud print printing control unit **802** of the cloud print server acquires transmission source information on the print request.

(130) In step **S1803**, the cloud print printing control unit **802** of the cloud print server changes the subsequent processing according to the transmission source information of the print request. In a case where the transmission source is a mobile terminal (YES in step **S1803**), the processing proceeds to step **S1808**. In step **S1808**, the cloud print job information management unit **806** of the cloud print server adds “job-release-method” to a job attribute of the target print job and sets “forced-print” as the value. In a case where the transmission source is not a mobile terminal (NO in step **S1803**), the processing proceeds to step **S1804**. In step **S1804**, the storage unit **805** of the cloud print server reads out the output method set for the target printing apparatus in the cloud printer management setting.

(131) In step **S1805**, the cloud print printing control unit **802** of the cloud print server changes the subsequent processing according to the output method.

(132) In a case where the output method is “forced store” (FORCED STORE in step **S1805**), the processing proceeds to step **S1806**. In step **S1806**, the cloud print printing control unit **802** of the cloud print server adds “job-release-method” to the job attribute of the target print job and sets “forced-store” as the value.

(133) In a case where the output method is “with printer setting” (WITH PRINTER SETTING in step **S1805**), the processing proceeds to step **S1807**. In step **S1807**, the cloud print printing control unit **802** of the cloud print server adds “job-release-method” to the job attribute of the target print job and sets “system-specified” as the value.

(134) In a case where the output method is “forced print” (FORCED PRINT in step **S1805**), the processing proceeds to step **S1808**. In step **S1808**, the cloud print printing control unit **802** of the cloud print server adds “job-release-method” to the job attribute of the target print job and sets “forced-print” as the value.

(135) In step **S1809**, the cloud print job information management unit **806** of the cloud print server sets “job-fetchable” as the value of the job attribute “job-state-reasons” of the target print job, whereby the print job can be acquired by the printing apparatus **101**.

Other Embodiments

(136) The present invention can also be implemented by processing in which a program for implementing one or more functions of the above-described embodiments is supplied to a system or an apparatus via a network or a storage, and one or more processors in a computer of the system or the apparatus read and execute the program. Further, the present invention can also be realized by a circuit (for example, an ASIC) that realizes one or more functions.

(137) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium.

The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(138) While the present invention has been described with reference to embodiments, it is to be understood that the invention is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(139) This application claims the benefit of Japanese Patent Application No. 2022-127523, filed Aug. 10, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A printing apparatus configured to receive a first print job from an external apparatus and print the first print job, the printing apparatus comprising: one or more processors; and at least one memory coupled to the one or more processors and having stored thereon instructions, which when executed by the one or more processors, cause the printing apparatus to: perform setting to enable a secure print setting with which the first print job received from the external apparatus is to be held without being printed, and the held first print job is printed in response to an instruction from a user to print the first print job; perform print control to receive a second print job from a print service via a network and print the second print job, wherein, in the print control, even in a case where the secure print setting is enabled, the second print job received from the print service is printed without holding the second print job in accordance with a job attribute of the received second print job; and receive a third print job from the print service, wherein, in a case where the print service determines whether a transmission source of a print request of the third print job stored in a virtual printer is a mobile terminal, wherein: if the transmission source is the mobile terminal, the print service specifies, as a value of the job attribute designating an output method of the third print job, a value indicating that printing is to be performed without holding the third print job in the printing apparatus, and provides the third print job to the printing apparatus; and if the transmission source is not the mobile terminal, the print service specifies, as the value of the job attribute designating the output method of the third print job, a value indicating that the third print job is to be held without being printed, and the held third print job is to be printed in response to an instruction from a user to print the third print job.
2. The printing apparatus according to claim 1, wherein the second print job received from the print service contains a job attribute specifying an output method of the second print job, and wherein, based on checking that a value of the job attribute is a value indicating that printing is to be performed without holding the second print job in the printing apparatus, the second print job is printed without holding the second print job.
3. The printing apparatus according to claim 2, wherein the job attribute specifying the output method of the second print job is a job attribute that is able to be specified by the print service, and wherein the first print job transmitted from the external apparatus is different from the second print job transmitted from the print service and does not contain the job attribute and the value.
4. The printing apparatus according to claim 2, wherein as the value of the job attribute specifying the output method of the second print job, at least a value indicating that printing is to be performed without holding the second print job in the printing apparatus, and a value indicating that the printing is to be performed with the secure print setting of the printing apparatus are able to be specified.
5. The printing apparatus according to claim 2, wherein based on checking that the value of the job attribute is a value indicating that the second print job is to be held in the printing apparatus without executing printing, the second print job is to be held until a print instruction is received

from a user.

6. A method for controlling a printing apparatus configured to receive a first print job from an external apparatus and print the first print job, the method comprising: performing setting to enable a secure print setting with which the first print job received from the external apparatus is to be held without being printed, and the held first print job is printed in response to an instruction from a user to print the first print job; performing print control to receive a second print job from a print service via a network and print the second print job, wherein, in the print control, even in a case where the secure print setting is enabled, the second print job received from the print service is printed without being held in accordance with a job attribute of the received second print job; and receiving a third print job from the print service, wherein the print service determines whether a transmission source of a print request of the third print job stored in a virtual printer is a mobile terminal, wherein: if the transmission source is the mobile terminal, the print service specifies, as a value of the job attribute designating an output method of the third print job, a value indicating that printing is to be performed without holding the third print job in the printing apparatus, and provides the third print job to the printing apparatus; and if the transmission source is not the mobile terminal, the print service specifies, as the value of the job attribute designating the output method of the third print job, a value indicating that the third print job is to be held without being printed, and the held third print job is to be printed in response to an instruction from a user to print the third print job.

7. The method according to claim 6, wherein the second print job received from the print service contains a job attribute specifying an output method of the second print job, and wherein, in the print control, based on checking that a value of the job attribute is a value indicating that printing is to be performed without holding the second print job in the printing apparatus, the second print job is printed without being held.

8. The method according to claim 7, wherein the job attribute specifying the output method of the second print job is a job attribute that is able to be specified by the print service, and wherein the first print job transmitted from the external apparatus is different from second print job transmitted from the print service and does not contain the job attribute and the value.

9. The method according to claim 7, wherein as the value of the job attribute specifying the output method of the second print job, at least a value indicating that printing is to be performed without holding the second print job in the printing apparatus, and a value indicating that the printing is to be performed with the secure print setting of the printing apparatus can be specified.

10. A non-transitory computer-readable storage medium storing executable instructions for executing a method for controlling a printing apparatus configured to receive a first print job from an external apparatus and print the first print job, the method comprising: performing setting to enable a secure print setting with which the first print job received from the external apparatus is to be held without being printed, and the held first print job is printed in response to an instruction from a user to print the first print job; performing print control to receive a second print job from a print service via a network and print the second print job, wherein, in the print control, even in a case where the secure print setting is enabled, the second print job received from the print service is printed without being held in accordance with a job attribute of the received second print job; and receiving a third print job from the print service, wherein the print service determines whether a transmission source of a print request of the third print job stored in a virtual printer is a mobile terminal, wherein: if the transmission source is the mobile terminal, the print service specifies, as a value of the job attribute designating an output method of the third print job, a value indicating that printing is to be performed without holding the third print job in the printing apparatus, and provides the third print job to the printing apparatus; and if the transmission source is not the mobile terminal, the print service specifies, as the value of the job attribute designating the output method of the third print job, a value indicating that the third print job is to be held without being

printed, and the held third print job is to be printed in response to an instruction from a user to print the third print job.
